

Assignment 3 – Part 1

Description

In the first part of the assignment, we are going to implement a fully generalized Attribute Interpolator that will allow to interpolate any attribute we have along the surface of the triangles we are going to draw the rest of the course.

Using this interpolator, we are going to implement an upgraded version of DrawTriangle, that will use advantage of our implemented Attribute Interpolator to simplify the code and allow for the using of any number of attributes when drawing our triangle.

Finally, we are going to implement three basic shaders that shall return the color interpolated, the inverse of the color interpolated and the blending of the color interpolated and the color already written in the frame buffer.

What to do:

You will have to implement the new methods in *AttributeInterpolator.h*, *FragmentShaders.h*, *DrawTriangle.h*:

AttributeInterpolator:

```
void Init(const float* vertices, const int attrCount);
```

```
bool Evaluate(float x, float y);
```

```
float* GetCurrentValues();
```

FragmentShaders:

```
Color FShaderColor(const FragmentInput& fragData);
```

```
Color FShaderColorInv(const FragmentInput& fragData);
```

```
Color FShaderAlphaBlend(const FragmentInput& fragData);
```

DrawTriangle:

```
void DrawTriangleShaded(const float* vertices, const unsigned int& vertexSize,  
ShaderFunc shader);
```

Submission Guidelines

Submit only AttributeInterpolator.cpp, FragmentShaders.cpp and DrawTriangle.cpp with a Readme.txt file as a zip folder on Moodle with the following naming conventions:

cs200fa25_{username}_3.zip

You will have until 23th of October to submit your code.

Additional guidelines

You have available in Moodle the following files:

- "*Code guidelines.pdf*": a complete file detailing different aspect of the code etiquette I expect you to follow.
- "*FileHeaderSample.h*": example of a complete and correct header for two function and a file.
- "*README_template.txt*": template of the readme file you have to include in your submission.

Some reminders about the submission:

- Be sure your code compiles.
- Double check no warnings are present in the compilation process.
- Don't forget to include proper documentation and headers.
- Naming convention is important!
- Use `Round()` when rounding, not `round()`, not `floor()` or anything else.